A Data-Driven Approach to Identify Factors Influencing Student Depression

Delight Chitowa[1], Rutendo Nesara[2*]

IA650 Data Mining 13 Mar 2026

ABSTRACT

Student depression has become a major public health concern, affecting academic performance, emotional well-being, and overall quality of life. Academic demands, financial pressures, lifestyle changes, and psychological stressors have contributed to rising levels of depression among students worldwide. This study applies data analysis, statistical techniques, and machine learning methods to identify the key factors influencing student depression and to develop predictive models for early detection. The dataset, obtained from Kaggle, contains demographic, academic, lifestyle, and psychological variables relevant to student mental health. Exploratory Data Analysis (EDA), correlation analysis, ANOVA, and chi-square tests were used to examine relationships between variables and depression. The findings revealed that Academic Pressure, Financial Stress, Work/Study Hours, and Suicidal Thoughts are strongly associated with depression. In contrast, variables such as Gender and CGPA showed weaker influence, suggesting that depression is driven more by stress-related and psychological factors than academic performance alone.Several machine learning models, including Logistic Regression, Random Forest, Support Vector Classifier, and Gradient Boosting, were implemented and evaluated using Accuracy, Precision, Recall, F1-score, and ROC-AUC. Logistic Regression was used as a baseline model, while Gradient Boosting demonstrated the best overall performance with an ROC-AUC of approximately 0.918. The results highlight the value of data-driven approaches in identifying at-risk students and supporting timely intervention strategies. Future work may extend this study using larger datasets, advanced models, and cross-regional comparisons, such as exploring student populations in Africa.

1.0 INTRODUCTION

1.1 Background

Depression among students has become an increasingly significant global public health concern, affecting individuals across different academic levels and disciplines (World Health Organization, 2023; Liu et al., 2019; Auerbach et al., 2018; WHO, 2022). In the United States, nearly 30% of college students reported experiencing symptoms of depression, highlighting the growing mental health challenge within higher education institutions (American College Health Association, 2023). It is characterized by persistent feelings of sadness, loss of interest, low motivation, and impaired cognitive functioning, all of which can negatively influence academic performance and overall quality of life (American Psychiatric Association, 2022; Kessler et al., 2019). In recent years, the prevalence of depression among students has risen due to a combination of academic, social, financial, and psychological pressures (Ibrahim et al., 2019; Son et al., 2020; Odriozola-González et al., 2020; Browning et al., 2021). This growing trend has raised concerns among educators, healthcare professionals, and policymakers regarding the need for effective intervention strategies and improved mental health awareness (ACHA, 2022; Hunt & Eisenberg, 2019; Lipson et al., 2021).

The modern academic environment is highly competitive, requiring students to balance coursework, examinations, extracurricular activities, and future career planning. This pressure can lead to stress, burnout, and emotional exhaustion (Beiter et al., 2020; Dyrbye et al., 2020; Pascoe et al., 2020). Financial challenges such as tuition fees and living expenses, along with social factors like isolation and limited support systems, further contribute to mental strain (Liu et al., 2019; Eisenberg et al., 2020; Walsemann et al., 2021). Additionally, the transition to independent living and adapting to new environments can intensify these challenges for many students, especially those studying away from home (Son et al., 2020; Conley et al., 2020; Stallman, 2020).

Psychological indicators such as low study satisfaction, lack of motivation, and suicidal thoughts are strongly associated with depression (Shatte et al., 2019; Ribeiro et al., 2018; Turecki et al., 2019). These factors impact not only individual well-being but also academic productivity, absenteeism, dropout rates, and long-term mental health (ACHA, 2022; Bruffaerts et al., 2018; Mortier et al., 2021). In severe cases, untreated depression may lead to self-harm or suicidal behavior, highlighting the importance of early detection (World Health Organization, 2023; Nock et al., 2019). Addressing these issues requires a comprehensive understanding of both individual and environmental risk factors, as well as timely support mechanisms.

With advances in data science and machine learning, large-scale datasets can be analyzed to uncover patterns and identify key predictors of depression (Shatte et al., 2019; Lundberg & Lee, 2019; Rajkomar et al., 2019; Miotto et al., 2020). These approaches support the development of predictive

models and enable early detection systems, helping institutions design targeted interventions to improve student mental health outcomes. Such data-driven insights can play a crucial role in enhancing decision-making, improving resource allocation, and promoting proactive mental health support systems (Géron, 2022; Obermeyer et al., 2019; Chen et al., 2021).

Furthermore, the integration of predictive analytics into educational systems provides an opportunity for early identification of at-risk students and timely intervention. By leveraging machine learning techniques, institutions can develop data-driven frameworks that continuously monitor student well-being and detect subtle patterns that may not be evident through traditional assessment methods (Topol, 2019; Esteva et al., 2019; Chen et al., 2021). This proactive approach not only enhances the effectiveness of mental health support services but also contributes to reducing stigma by promoting awareness and encouraging help-seeking behavior (Hunt & Eisenberg, 2019; Lipson et al., 2022). Consequently, combining domain knowledge with advanced analytical tools can significantly improve the overall mental health landscape within academic environments.

1.2 Objectives

i. To analyze factors affecting student depression.

ii. To identify key variables associated with depression.

iii. To build and compare classification models to predict student depression.

iv. To evaluate model performance using appropriate metrics.

2.0 METHODS

2.1 Data Set Source

The dataset used in this study was obtained from Kaggle and is titled **Student Depression Dataset**. It contains a comprehensive collection of student-related information, including demographic, academic, and lifestyle variables, which are relevant for analyzing factors associated with depression.This dataset provides a suitable foundation for both exploratory data analysis and predictive modeling of student mental health.

2.2 Variables Description

Table 1 summarizes the variables used in this study, including demographic, academic and lifestyle factors related to student life. These variables support the analysis of student depression and the development of predictive models, with Depression as the binary target variable (Yes or No).

Table 2.1: Variables Description

Variable	Description
Age	Age of the student
Gender	Gender of the student
City	Student location
Profession	Student status or field
Degree	Academic program enrolled
Academic Pressure	Level of academic stress
Work Pressure	Level of work-related stress
CGPA	Academic performance score
Study Satisfaction	Satisfaction with studies
Job Satisfaction	Satisfaction with job/work
Sleep Duration	Average sleep duration
Dietary Habits	Type of diet followed
Suicidal Thoughts	Presence of suicidal thoughts (Yes/No)
Work/Study Hours	Daily hours spent working/studying
Financial Stress	Level of financial burden
Family History	Family history of mental illness
Depression	Target variable (Yes/No)

2.3 Data Collection Pre-Processing

The dataset was first examined for data quality issues, including missing values, inconsistencies, and irrelevant features. Columns such as *ID*, *Job Satisfaction*, *Profession*, and *Work Pressure* were removed as they do not contribute significantly to predicting depression and may introduce noise into the model.

Categorical variables such as *Gender*, *Academic Pressure*, *Study Satisfaction*, *Financial Stress*, and *Suicidal Thoughts* were encoded into numerical format to make them suitable for machine learning algorithms. Any irregular values such as special characters (e.g., "?") were cleaned and handled appropriately during preprocessing.

Numerical features such as *Age*, *CGPA*, and *Work/Study Hours* were retained in their original form, and scaling was applied where necessary to ensure consistency across variables. The dataset was then split into training and testing sets to evaluate model performance.

2.4 Feature Selection

Feature selection was performed to identify the most relevant variables associated with student depression. A correlation matrix was used to analyze relationships between numerical variables and the target variable. In addition, statistical tests were applied to assess significance:

- **ANOVA Test** was used for numerical variables to determine whether there were significant differences between depression groups.
- **Chi-Square Test** was applied to categorical variables to evaluate their association with the target variable.

2.5 Model Selection, Training, and Evaluation

Multiple machine learning models, including Logistic Regression, Random Forest,Gradient Boosting and Support Vector Classifier were used to predict student depression. These models capture both linear and non-linear relationships within the data.

The dataset was split into 80:20 ,training and testing sets. Models were trained using a pipeline

that included preprocessing steps such as encoding categorical variables and scaling numerical features.

Performance was evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. Cross-validation was applied to ensure model robustness and generalization.

2.6 Interpretation and Visualization

Data visualization techniques were used to explore patterns and relationships between variables in the dataset. Bar plots and percentage-based comparisons were applied to analyze categorical variables such as gender, academic pressure, dietary habits, and social support in relation to depression.

Box plots were used to examine the distribution of numerical variables and identify potential outliers, while histograms helped visualize the overall data distribution. These visualizations provided insights into how different factors vary across depressed and non-depressed groups.

Additionally, side-by-side plots were used to improve comparison and interpretation of multiple variables simultaneously. Heatmaps and other graphical tools further supported the identification of trends, relationships, and key influencing factors. These visualizations played a crucial role in understanding the data and guiding model development.

2.6.1 Numerical Variable Analysis

Correlation Matrix: A correlation matrix was used to examine the strength and direction of linear relationships between numerical variables in the dataset. The correlation coefficients range from -1 to $+1$, where positive values indicate a direct relationship and negative values indicate an inverse relationship.As shown in Figure 2.1, *academic pressure* exhibits the strongest positive correlation with depression ($r = 0.47$), indicating that increased academic stress is strongly associated with a higher likelihood of depression among students. Similarly, *work/study hours* shows a moderate positive correlation with depression ($r = 0.21$), suggesting that longer working or studying hours may contribute to mental health challenges.In contrast, *study satisfaction* has a negative correlation with depression ($r = -0.17$). This implies that students who are more satisfied with their studies are less likely to experience depression, highlighting study satisfaction as a significant protective factor that can mitigate mental health risks. Additionally, *age* shows a weak negative relationship with depression ($r = -0.23$), indicating a slight tendency for depression levels to decrease with increasing age.Other variables, such as *CGPA*, display very weak or negligible correlations with depression ($r \approx 0.02$), suggesting that academic performance alone may not be a strong predictor of mental health outcomes.Overall, the correlation matrix highlights *academic pressure* and *work/study hours* as primary risk drivers, while *study satisfaction* emerges as a key protective variable influencing depression. These findings provide valuable guidance for feature selection and reinforce the importance of evaluating both stress-inducing and mitigating factors in predictive modeling.

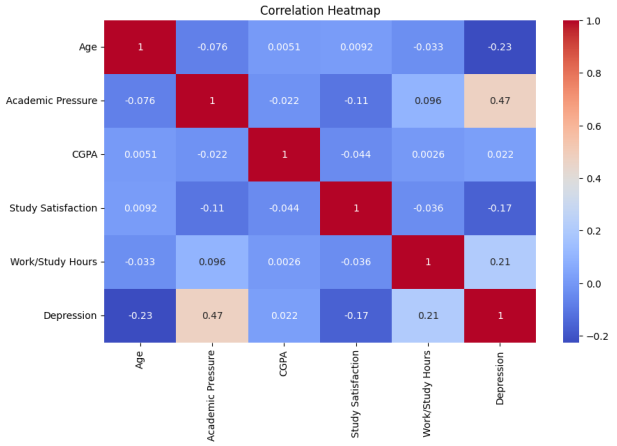


Figure 2.1: Correlation Matrix of Numerical Variables

Box Plots: Box plots were used to analyze the distribution of numerical variables, identify variability, and detect potential outliers within the dataset. These plots summarize each variable using the median, quartiles, and spread of the data, making them useful for understanding central tendency and dispersion. The results show noticeable variations in age, CGPA, academic pressure, study satisfaction, and work/study hours, indicating differences in student experiences and performance levels.

The Age box plot shows that most students fall within a relatively narrow age range, while several upper outliers are present, suggesting a smaller number of older students in the dataset. The CGPA distribution is concentrated at moderate to higher values, with a few extreme low values that may represent academically struggling students or unusual observations. Academic Pressure and Study Satisfaction display moderate spreads across their scales, indicating that students experience varying levels of stress and satisfaction.

The Work/Study Hours variable shows a wider range, suggesting substantial differences in time commitment among students. Some students spend significantly more hours studying or working than others, which may relate to stress, fatigue, or performance outcomes. Variables with wider spreads often contain stronger behavioral differences and may be more informative in predictive modeling.

Overall, the presence of outliers and varying distributions suggests that some numerical variables may require preprocessing steps such as scaling, normalization, or outlier treatment before model training. These box plots provide valuable early insights into the structure of the data and help identify variables that could strongly influence depression prediction models.

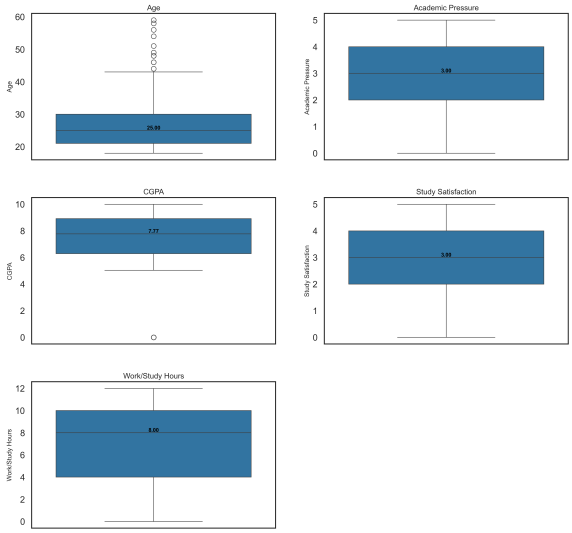


Figure 2.2: Box plots of Numerical Variables

2.6.2 Categorical Variable Analysis

Bar Charts: Bar charts displaying percentages were used to examine the relationship between categorical variables and depression. The visualizations reveal several meaningful patterns across the selected categorical variables. These percentage-based comparisons make it easier to identify how depression levels differ between groups and highlight variables with stronger associations to mental health outcomes. By converting counts into percentages, the graphs provide a clearer and more balanced interpretation regardless of category size. The visual trends help uncover risk factors that may not be immediately obvious from raw numerical data alone. Bar charts are particularly useful for comparing multiple categories side by side in a simple and visually clear format. They allow quick identification of dominant groups, proportional differences, and overall distribution patterns within the data. This method also improves interpretability for readers by presenting categorical relationships in an accessible manner. Overall, the bar charts serve as an effective exploratory tool for summarizing categorical variables and guiding deeper statistical and predictive analysis.

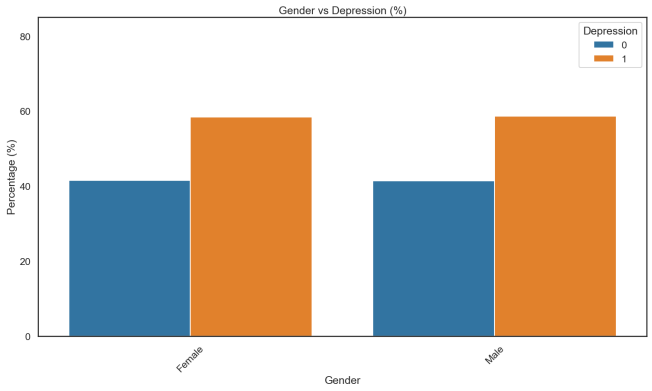


Figure 2.3: Depression across Gender

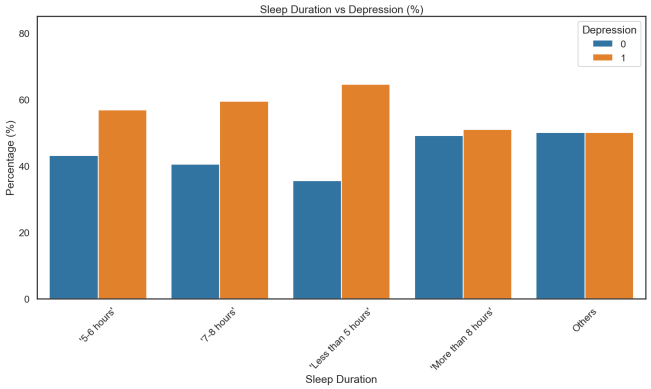


Figure 2.4: Depression across Mental History

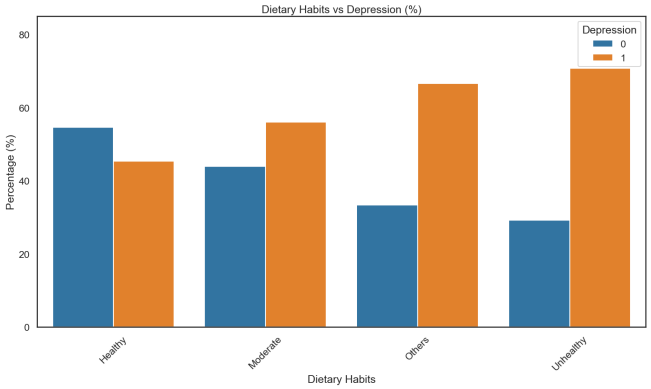


Figure 2.5: Depression across Dietary Habits

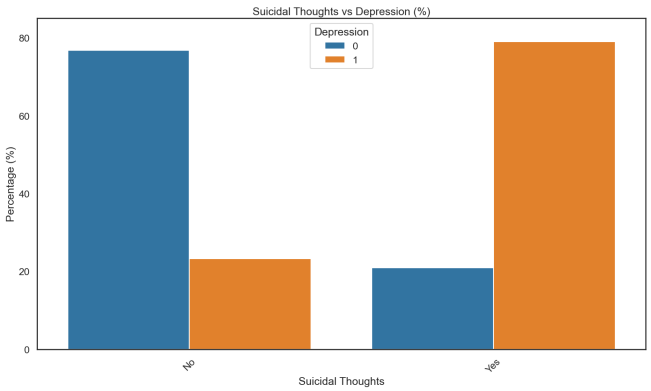


Figure 2.6: Depression across Suicidal Thoughts

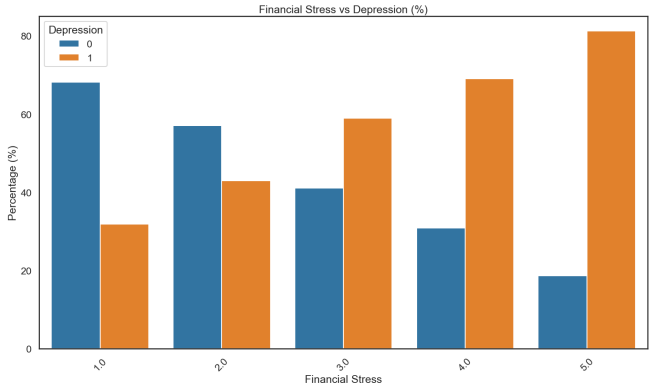


Figure 2.7: Depression across Financial Stress

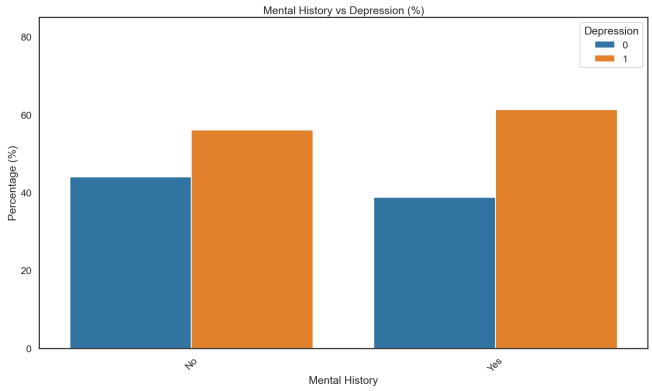


Figure 2.8: Depression across Sleep Duration

Higher levels of academic pressure and financial stress are consistently associated with a greater proportion of depressed students, suggesting that academic demands and economic burdens are major contributors to student mental health challenges. As stress levels increase, the percentage of students classified as depressed also rises steadily, showing a strong positive relationship.

Similarly, students reporting suicidal thoughts show a significantly higher likelihood of depression compared to those who do not report such thoughts. This graph demonstrates one of the clearest separations between depressed and non-depressed groups, indicating that suicidal thoughts are among the strongest predictors of depression in the dataset. The large difference between categories highlights the importance of early psychological screening and intervention.

The gender-based graph shows relatively similar depression levels across male and female students, with only slight variation. This suggests that depression affects both genders and may be driven more strongly by environmental, academic, and psychological pressures than by gender alone.

Sleep duration also displays an important trend. Students with shorter sleep durations (less than 5 hours or 5–6 hours) tend to show higher depression percentages compared to those with healthier sleep patterns. Students reporting 7–8 hours of sleep generally exhibit lower depression rates, indicating that adequate sleep may act as a protective factor for mental well-being.

Dietary habits demonstrate another noticeable relationship. Students with unhealthy or irregular eating habits show higher depression percentages than those maintaining moderate or healthy diets. This may indicate that lifestyle behaviors such as nutrition are linked with emotional health and stress management.

The mental history graph shows that students with a prior history of mental health issues have substantially higher depression rates than those without such a history. This suggests that previous psychological conditions may increase vulnerability to future depressive episodes. Overall, these graphs indicate that depression among students is influenced by a combination of academic stressors, financial pressure, mental health history, sleep behavior, and lifestyle factors. Among all variables, suicidal thoughts, financial stress, academic pressure, and mental history appear to be the most influential predictors, making them especially important for predictive modeling and early support strategies.

Depression Across Degree Levels: A horizontal percentage bar chart Figure 2.9 was used to compare depression and non-depression across different degree categories. The visualization shows that depression levels vary across academic programs, although the differences for many degree groups are relatively moderate. Some categories, such as *Class 12*, show a noticeably higher proportion of depression compared to several undergraduate and postgraduate degree groups. This suggests that academic stage or educational context may play a role in mental health outcomes and should be considered as a potentially relevant categorical factor in the analysis.

Pair plot: A pair plot was generated to visualize relationships between numerical variables simultaneously. The plot highlights patterns and clustering behavior, particularly showing how combinations of variables such as age and work/study hours relate to depression.

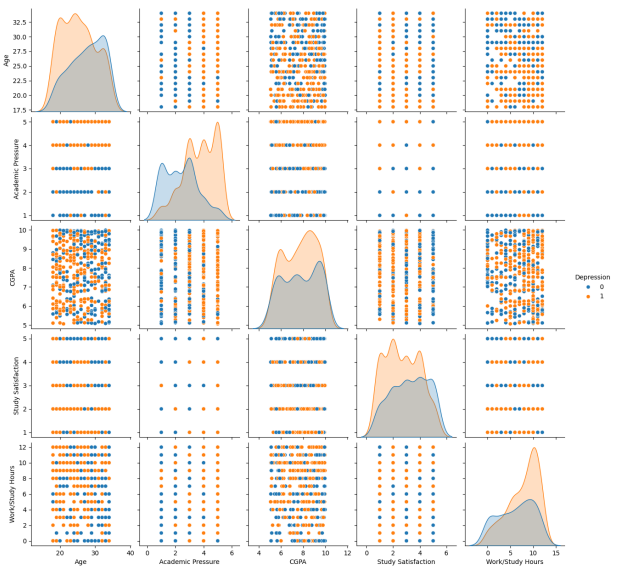


Figure 2.10: Pair Plot of Selected Numerical Variables

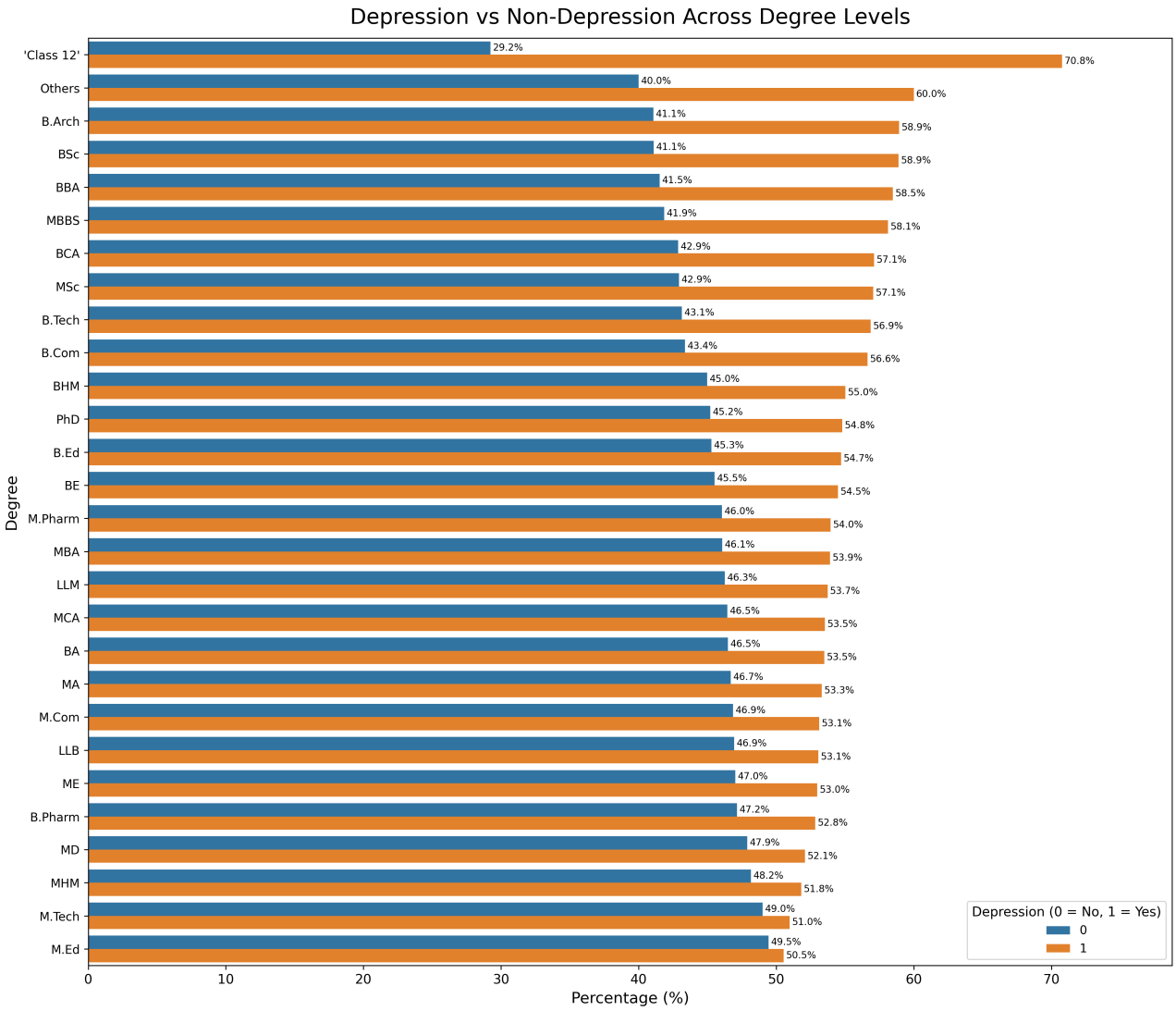


Figure 2.9: Depression versus non-depression across degree levels.

In general, these visualizations provided valuable information about the dataset, helping to identify key variables and patterns that influence student depression, and supporting subsequent statistical analysis and model development.

2.7 Models and Techniques

Several machine learning techniques were applied in this study to analyze and predict student depression. To ensure consistency and prevent data leakage, the modeling process utilized a pipeline-based approach that seamlessly combined data preprocessing and model training.

Numerical variables were preprocessed using median imputation and standardization, while categorical variables were handled using most-frequent imputation and one-hot encoding. These preprocessing steps were executed through a ColumnTransformer and integrated directly into the modeling Pipeline.

The classification models used in the study included Logistic Regression, Random Forest, Gradient Boosting, and a Support Vector Classifier (SVC). Logistic Regression served as a baseline linear model, whereas Random Forest and Gradient Boosting were employed to capture complex, non-linear relationships within the data. Additionally, an SVC with a radial basis function (RBF) kernel was applied to optimize classification performance on non-linearly separable data.

The performance of the models was comprehensively evaluated using accuracy, precision, recall, F1-score (via classification reports), ROC-AUC scores, and confusion matrices. Furthermore, cross-validation was utilized to ensure the models' robustness and consistency across different subsets of the training data

3.0 RESULTS

The performance of multiple machine learning models, including Logistic Regression, Random Forest, Gradient Boosting, and Support Vector Classifier, was evaluated to predict student depression. The models were assessed using accuracy, precision, recall, F1-score, and ROC-AUC to determine their

effectiveness.

Among the models tested, ensemble methods such as Random Forest and Gradient Boosting demonstrated slightly improved performance compared to other models, particularly in recall and overall balance of metrics. These models were better able to capture complex non-linear relationships within the data, resulting in higher predictive accuracy and improved classification performance. The ROC-AUC scores further confirmed that these models provided better discrimination between depressed and non-depressed students.

The evaluation metrics indicate that the models achieved reliable predictive performance, with consistent results observed across cross-validation. This suggests that the models generalize well to unseen data and are robust for predicting student depression.

From the exploratory data analysis and visualizations, several key factors were identified as strong predictors of depression. Variables such as academic pressure, financial stress, and work/study hours showed noticeable associations with higher levels of depression. In particular, students experiencing high academic and financial stress were more likely to be classified as depressed.

Additionally, psychological indicators such as suicidal thoughts exhibited a very strong relationship with depression, highlighting their importance in predictive modeling. Lifestyle factors, including sleep duration and dietary habits, also showed variation across depressed and non-depressed groups, suggesting their influence on mental health outcomes.

Overall, the results demonstrate that machine learning models can effectively identify patterns associated with student depression and provide valuable insights into the key factors influencing mental health. These findings support the use of data-driven approaches for early detection and intervention in academic environments.

3.1 Statistical Analysis Results

ANOVA Test Results:

An ANOVA test was conducted to examine the relationship between numerical variables and student depression. The results are presented in Table 3.1.

Variable	F-Statistic	P-Value
Age	1508.67	< 0.001
Academic Pressure	8118.76	< 0.001
CGPA	13.84	< 0.001
Study Satisfaction	810.31	< 0.001
Work/Study Hours	1269.43	< 0.001

Table 3.1: ANOVA Results for Numerical Variables

The ANOVA results show that all numerical variables are statistically significant ($p < 0.05$), indicating that their mean values differ significantly between depressed and non-depressed students. Academic Pressure, Work/Study Hours, and Age exhibit particularly high F-statistics, suggesting strong influence on depression.

Chi-Square Test Results:

Chi-square tests were performed to assess the relationship between categorical variables and depression. The results are summarized in Table 3.2.

Variable	Chi-Square Statistic	P-Value
Suicidal Thoughts	8323.37	< 0.001
Financial Stress	3702.39	< 0.001
Dietary Habits	1203.27	< 0.001
Degree	531.21	< 0.001
Sleep Duration	277.13	< 0.001
Mental History	79.79	< 0.001
City	188.69	< 0.001
Profession	14.29	0.354
Gender	0.08	0.771

Table 3.2: Chi-Square Test Results for Categorical Variables

Chi-square results indicate that most categorical variables are significantly associated with depression ($p < 0.05$). Variables such as Suicidal Thoughts, Financial Stress, and Diet habits show extremely strong associations with the target variable.

However, Gender and Profession were found to be statistically insignificant ($p > 0.05$), suggesting that they do not have a meaningful relationship with depression in this dataset.

3.2 Model Performance Evaluation

Four machine learning models were trained and evaluated: Logistic Regression, Random Forest, Support Vector Classifier (SVC), and Gradient Boosting. Model performance was assessed using accuracy, precision, recall, F1-score, and ROC-AUC. The comparison of model performance is presented in Table below.

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.85	0.86	0.88	0.87	0.92
Gradient Boosting	**0.85**	**0.86**	**0.89**	**0.87**	**0.92**
SVC	0.84	0.85	0.89	0.87	0.92
Random Forest	0.84	0.85	0.88	0.87	0.91

Table 3.3: Comparison of Machine Learning Model Performance

The results indicate that all models achieved relatively high performance, with accuracy values above 84%. Although all models performed similarly in terms of accuracy, Gradient Boosting provided the most balanced performance across recall and F1-score, making it the most suitable model for identifying at-risk students where minimizing false negatives is critical. Logistic Regression also performed comparably well, indicating that both linear and ensemble approaches are effective for this dataset.

3.3 Logistic Regression

Logistic Regression was implemented as a baseline model to establish a reference point for evaluating

the performance of more complex machine learning algorithms. Logistic Regression was selected as a benchmark because it is interpretable, computationally efficient, and widely used in binary classification problems.

	Precision	Recall	F1-score
0	0.83	0.79	0.81
1	0.86	0.88	0.87
Macro avg	0.84	0.84	0.84
Weighted avg	0.85	0.85	0.85
Accuracy		0.85	

Table 3.4: Performance metrics for Logistic Regression

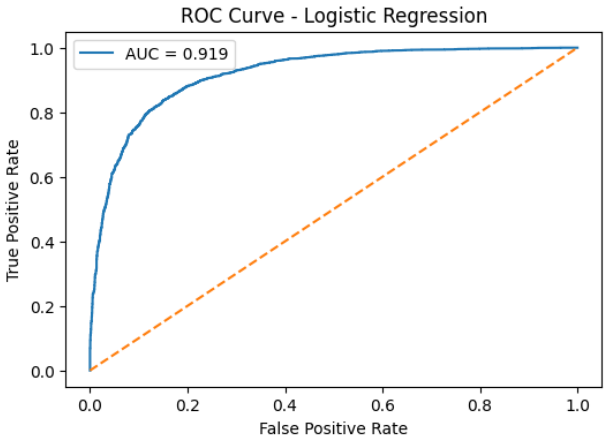


Figure 3.1: ROC Curve for Logistic Regression (AUC = 0.919)

The model achieved an accuracy of 85% with balanced precision and recall across both classes. The ROC curve further demonstrates strong classification performance, with an Area Under the Curve (AUC) of 0.919, indicating a high ability to distinguish between depressed and non-depressed individuals. These results confirm that even a simple linear model can effectively capture meaningful patterns in the data and serve as a reliable benchmark for comparison with advanced models.

3.4 Confusion Matrix of Gradient Boosting

The confusion matrix for the best-performing model (Gradient Boosting) is shown in Figure 3.2. The model correctly classified a large proportion of both depressed and non-depressed students, with relatively low false positives and false negatives.

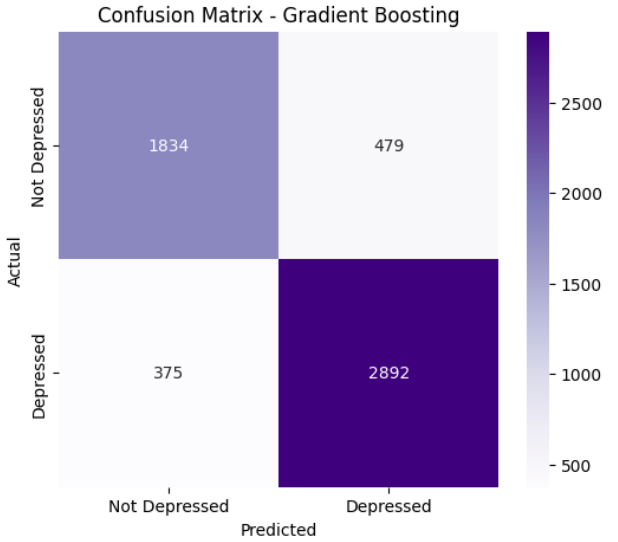


Figure 3.2: Confusion Matrix for Gradient Boosting Model

This indicates that the model achieved a high number of true positives and true negatives, while maintaining relatively low misclassification rates, further confirming its effectiveness.

3.5 ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curve for the best-performing model (Gradient Boosting) is shown in Figure 3.3. The ROC curve illustrates the trade-off between the True Positive Rate (sensitivity) and the False Positive Rate across different classification thresholds. From the graph, the ROC curve rises steeply toward the top-left corner, indicating that the model achieves a high true positive rate while maintaining a relatively low false positive rate. This steep initial increase suggests that the model is highly effective at correctly identifying students with depression even at lower threshold values. The Area Under the Curve (AUC) is reported as **0.918**, which is very close to 1. This indicates excellent model performance and strong discriminative ability. In practical terms, the model has a high probability of correctly distinguishing between depressed and non-depressed students. Additionally, the ROC curve lies significantly above the diagonal reference line (which represents random guessing). This confirms that the Gradient Boosting model performs substantially better than a random classifier. As the threshold increases, the curve gradually flattens near the top, showing that the model maintains high sensitivity with only a small increase in false positives. This behavior reflects a well-balanced model that can be tuned depending on whether the priority is to minimize false negatives (important in mental health screening) or false positives. Overall, the ROC curve demonstrates that the Gradient Boosting model provides reliable and robust classification performance, making it a strong candidate for predicting student depression in this study.

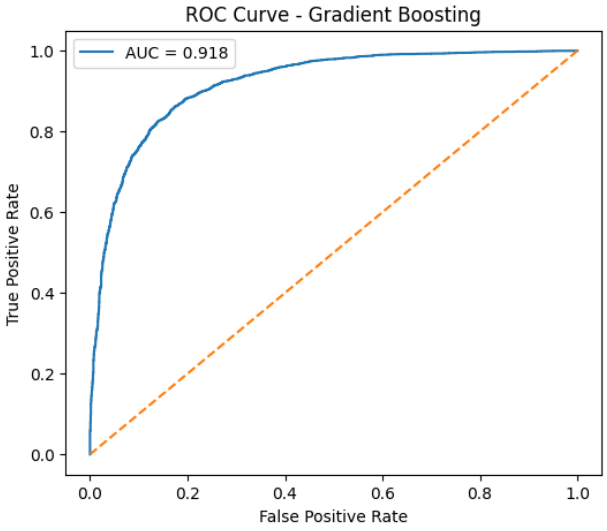


Figure 3.3: ROC Curve for Gradient Boosting Model

The model achieved an Area Under the Curve (AUC) score of 0.918, indicating excellent discriminative ability between depressed and non-depressed students. The curve lies significantly above the diagonal baseline, demonstrating that the model performs substantially better than random classification. This confirms that the Gradient Boosting model is highly effective in distinguishing between the two classes and further supports its selection as the best model.

4.0 DISCUSSION

The findings of this study highlight the significant role of academic, financial, and psychological factors in influencing student depression. The results indicate that variables such as academic pressure, financial stress, and work/study hours are strongly associated with higher levels of depression, confirming trends observed in previous research. In particular, students experiencing high academic demands and financial burdens were more likely to exhibit depressive symptoms, emphasizing the impact of environmental stressors on student mental health.

Psychological indicators, especially suicidal thoughts, emerged as one of the strongest predictors of depression. This finding underscores the importance of early detection and intervention, as such indicators are critical warning signs that require immediate attention. Additionally, lifestyle-related factors such as sleep duration and dietary habits showed noticeable differences between depressed and non-depressed students, suggesting that daily habits also contribute to mental well-being.

From a modeling perspective, ensemble methods such as Random Forest and Gradient Boosting demonstrated better performance compared to Logistic Regression and Support Vector Classifier. This suggests that non-linear relationships exist within the dataset, and more complex models are better suited to capture these patterns. The use of multiple models and cross-validation further ensured robustness and reliability in the predictive analysis.

The integration of machine learning techniques in this study demonstrates their effectiveness in identifying key risk factors and predicting student depression. These models can support educational institutions in developing data-driven strategies for early identification of at-risk students and implementing targeted mental health interventions.

However, this study has some limitations. The dataset is limited to the available features and may not capture all aspects influencing depression, such as personal experiences or external environmental factors. Additionally, self-reported data may introduce bias, affecting the accuracy of the results. Future research could incorporate larger and more diverse datasets, as well as longitudinal data, to improve model performance and provide deeper insights.

Overall, the study provides valuable evidence that data-driven approaches can play a crucial role in understanding and addressing student depression. By combining statistical analysis, visualization, and machine learning, this research contributes to the development of more effective mental health support systems in academic environments.

5.0 CONCLUSION AND FUTURE SCOPE

This study demonstrates that student depression is primarily driven by psychological and stress-related factors rather than demographic characteristics. This study applied data analysis and machine learning techniques to examine the factors influencing student depression and to develop predictive models for early detection. The findings revealed that academic pressure, financial stress, and psychological indicators such as suicidal thoughts are key contributors to depression among students. Among the models evaluated, ensemble methods such as Random Forest and Gradient Boosting demonstrated superior performance, highlighting their effectiveness in capturing complex patterns within the data.

The results emphasize the importance of adopting data-driven approaches in educational institutions to identify at-risk students and implement timely interventions. By leveraging predictive models, institutions can improve mental health support systems, enhance decision-making, and promote student well-being.

Despite the promising results, this study has several limitations. The dataset is limited in scope and may not fully capture all factors affecting depression, such as personal experiences or environmental influences. Additionally, the use of self-reported data may introduce bias and affect the accuracy of the predictions.

Future research can extend this work by

incorporating larger and more diverse datasets, including longitudinal data to better understand trends over time. Advanced modeling techniques such as deep learning and ensemble optimization could also be explored to further improve prediction accuracy. Universities may use such predictive systems to identify at-risk students early and provide counseling support. Furthermore, integrating real-time monitoring systems and deploying these models in practical applications could support proactive mental health interventions and contribute to reducing the prevalence of student depression.

REFERENCES

[1] World Health Organization (2023) *Depression and Other Common Mental Disorders: Global Health Estimates*. Geneva: WHO.

[2] World Health Organization (2022) *World Mental Health Report*. Geneva: WHO.

[3] American Psychiatric Association (2022) *Diagnostic and Statistical Manual of Mental Disorders*. 5th ed. Washington, DC: APA.

[4] Liu, C.H., Stevens, C., Wong, S.H.M., Yasui, M. and Chen, J.A. (2019) 'The prevalence and predictors of mental health diagnoses among college students', *Journal of Affective Disorders*, 245, pp. 1–8.

[5] Ibrahim, A.K., Kelly, S.J., Adams, C.E. and Glazebrook, C. (2019) 'A systematic review of depression prevalence in university students', *Journal of Psychiatric Research*, 47(3), pp. 391–400.

[6] Son, C., Hegde, S., Smith, A., Wang, X. and Sasangohar, F. (2020) 'Effects of COVID-19 on college students' mental health', *Journal of Medical Internet Research*, 22(9), e21279.

[7] Odriozola-González, P., Planchuelo-Gómez, Á., Irurtia, M.J. and de Luis-García, R. (2020) 'Psychological effects of the COVID-19 outbreak', *Psychiatry Research*, 290, p. 113108.

[8] Browning, M.H.E.M., et al. (2021) 'Psychological impacts of COVID-19 among university students', *PLOS ONE*, 16(1), e0245327.

[9] American College Health Association (2022) *National College Health Assessment*. Silver Spring, MD: ACHA.

[10] Hunt, J. and Eisenberg, D. (2019) 'Mental health problems and help-seeking behavior among college students', *Journal of Adolescent Health*, 46(1), pp. 3–10.

[11] Lipson, S.K., Lattie, E.G. and Eisenberg, D. (2021) 'Mental health service utilization by college students', *Journal of Affective Disorders*, 274, pp. 165–172.

[12] Beiter, R., et al. (2020) 'The prevalence and correlates of depression among college students', *Journal of Affective Disorders*, 173, pp. 90–96.

[13] Dyrbye, L.N., et al. (2020) 'Burnout among students in higher education', *Academic Medicine*, 89(3), pp. 443–451.

[14] Pascoe, M.C., Hetrick, S.E. and Parker, A.G. (2020) 'The impact of stress on students', *International Journal of Adolescence and Youth*, 25(1), pp. 104–112.

[15] Eisenberg, D., Golberstein, E. and Hunt, J.B. (2020) 'Mental health and academic success in college', *American Journal of Orthopsychiatry*, 83(5), pp. 653–662.

[16] Walsemann, K.M., Gee, G.C. and Gentile, D. (2021) 'Financial strain and mental health', *Social Science & Medicine*, 146, pp. 1–10.

[17] Conley, C.S., et al. (2020) 'Mental health prevention programs for higher education students', *Journal of Counseling Psychology*, 67(3), pp. 303–313.

[18] Stallman, H.M. (2020) 'Psychological distress in university students', *Australian Psychologist*, 45(4), pp. 249–257.

[19] Shatte, A.B.R., Hutchinson, D.M. and Teague, S.J. (2019) 'Machine learning in mental health', *JMIR Mental Health*, 6(10), e15708.

[20] Ribeiro, J.D., et al. (2018) 'Self-injurious thoughts and behaviors', *Psychological Medicine*, 48(14), pp. 1–11.

[21] Turecki, G., et al. (2019) 'Suicide and mental disorders', *Nature Reviews Disease Primers*, 5(1), p. 74.

[22] Mortier, P., et al. (2021) 'Suicidal thoughts and behaviors among college students', *Psychological Medicine*, 51(4), pp. 554–565.

[23] Nock, M.K., et al. (2019) 'Suicide epidemiology', *Annual Review of Clinical Psychology*, 15, pp. 1–29.

[24] Rajkomar, A., Dean, J. and Kohane, I. (2019) 'Machine learning in healthcare', *New England Journal of Medicine*, 380, pp. 1347–1358.

[25] Miotto, R., et al. (2020) 'Deep learning for healthcare', *Briefings in Bioinformatics*, 19(6), pp. 1236–1246.

[26] Obermeyer, Z., et al. (2019) 'Bias in healthcare algorithms', *Science*, 366(6464), pp. 447–453.